CS 58500 – Theoretical Computer Science Toolkit

Lecture 14 (03/24)

Markov Chains

https://ruizhezhang.com/course_spring_2026.html

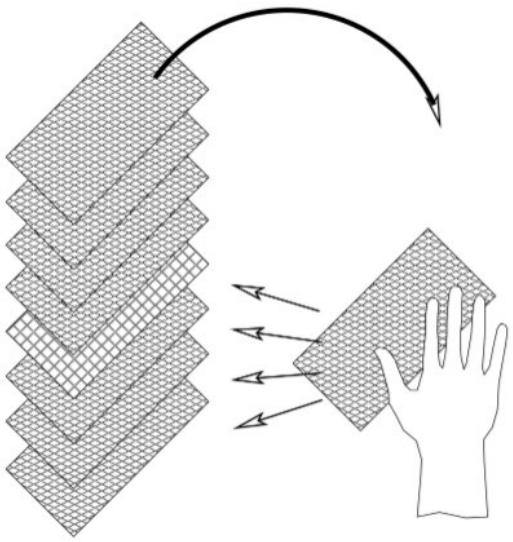
Today’s Lecture

- Introduction to Markov Chains
- Speed of Convergence
- Spectral Theory of Markov Chains
- Poincaré Inequality

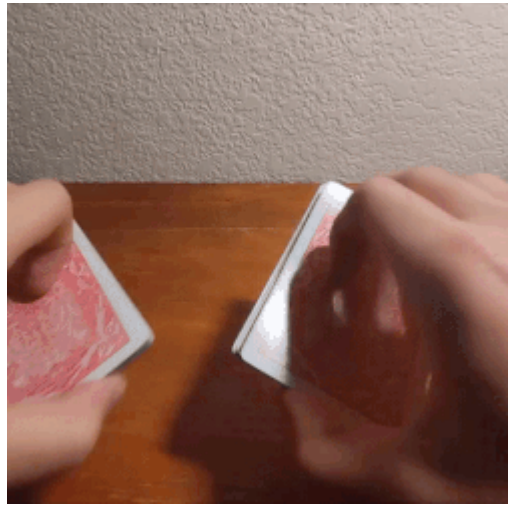
Card Shuffling

How many times do you need to shuffle a new deck of cards

Top-to-Random Shuffle



Riffle Shuffle



- Choose $k \sim \mathrm{Bin}\left(n, \frac{1}{2}\right)$
- $L \coloneqq$ top k cards
- $R \coloneqq$ bottom $n - k$ cards
- Drop the cards at random from L or R one card at a time
- Choose L with probability $\frac{\ell}{\ell+r}$, where ℓ, r are the sizes of L, R at that time
- Choose R with probability $\frac{r}{\ell+r}$

Card Shuffling

How many times do you need to shuffle a new deck of cards

- Mathematically, we can consider the order of the cards as a permutation $\sigma: [n] \to [n]$
- A new deck of cards (unshuffled) corresponds to $\sigma = \mathrm{id} \coloneqq \begin{pmatrix} 1 & 2 & \cdots & n \\ 1 & 2 & \cdots & n \end{pmatrix}$
- A fully shuffled deck corresponds to a uniformly random permutation from $\mathcal{S}_n$, the set (group) of all $n!$ permutations of $[n]$
- Each iteration of shuffling can be viewed as randomly "walking" from current permutation σ to a new permutation $\sigma' \in \mathcal{S}_n$

Fact. Top-to-Random Shuffle needs $\mathcal{O}(n \ln n)$ iterations, and Riffle Shuffle needs $\mathcal{O}(\ln n)$ iterations. When $n = 52$, seven times of Riffle Shuffle are sufficient

Markov Chains

A (discrete-time) Markov chain on Ω is a sequence of random variables $\{X_t\}_{t=0}^{\infty}$ taking values in Ω satisfying the Markov property:

$$\Pr[X_t = x_t | X_0 = x_0, \dots, X_{t-1} = x_{t-1}] = \Pr[X_t = x_t | X_{t-1} = x_{t-1}] \quad \forall t \geq 0, x_0, \dots, x_t \in \Omega$$

- Transition probability matrix (or Markov kernel) $\boldsymbol{P} \in \mathbb{R}_{\geq 0}^{\Omega \times \Omega}$:

$$\boldsymbol{P}(x, y) = \boldsymbol{P}(x \to y) \coloneqq \Pr[X_t = y | X_{t-1} = x] \quad \forall x, y \in \Omega$$

  $\boldsymbol{P}(x \to \cdot)$ (i.e., the x-th row of $\boldsymbol{P}$) is a probability distribution over Ω

- Initial distribution $\mu_0 \in \mathbb{R}_{\geq 0}^{\Omega}$ specifying the law of X_0
- Let μ_t be the law of X_t. Then we have

$$\mu_t = \mu_0 \boldsymbol{P}^t$$

Markov Chains

Random walk on graphs

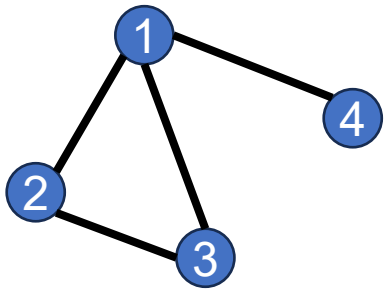
Let $G = (V, E)$ be an undirected graph. The simple random walk on G is a Markov chain with state space $\Omega = V$ described by the following process:

If the current vertex is $u \in V$, then we transition to a uniformly random neighbor of u.

- Transition probability matrix:

$$\boldsymbol{P}_G = \boldsymbol{D}_G^{-1} \boldsymbol{A}_G$$

where $\boldsymbol{A}_G$ denotes the adjacency matrix of G, and $\boldsymbol{D}_G = \mathbf{Diag}(\deg_G(v) : v \in V)$ is the diagonal degree matrix for normalization



$$\boldsymbol{A}_G = \begin{bmatrix} & 1 & 1 & 1 \\ 1 & & 1 & \\ 1 & 1 & & \\ 1 & & & \end{bmatrix} \quad \boldsymbol{D}_G = \begin{bmatrix} 3 & & & \\ & 2 & & \\ & & 2 & \\ & & & 1 \end{bmatrix} \quad \boldsymbol{P}_G = \begin{bmatrix} & 1/3 & 1/3 & 1/3 \\ 1/2 & & 1/2 & \\ 1/2 & 1/2 & & \\ 1 & & & \end{bmatrix}$$

Markov Chains

We say a Markov chain P is reversible w.r.t. a distribution μ if they satisfy the detailed balance condition:

$$\mu(x) \cdot \boldsymbol{P}(x \to y) = \mu(y) \cdot \boldsymbol{P}(y \to x) \qquad \forall x, y \in \Omega$$

- If P is symmetric, i.e., $P(x \to y) = P(y \to x)$, then P is reversible w.r.t. the uniform distribution over Ω
- The simple random walk on a graph G is reversible w.r.t. $\mu(v) \propto \deg_G(v)$
- Every reversible Markov chain corresponds to a random walk on a weighted, undirected graph $G = (\Omega, E)$: we connect $x \sim y$ if $P(x \to y) > 0$ and we assign weight $\mu(x)P(x \to y)$ to this edge
- $\mu(x)P(x \to y)$ is called the ergodic flow from x to y

Detailed balance condition $\leftrightsquigarrow$ "Flow in = Flow out"

Markov Chains

A probability distribution μ on Ω is stationary with respect to a Markov chain P if $\mu \boldsymbol{P} = \mu$

- For the random walk on a graph G, the stationary distribution $\mu_G(v) \propto \deg_G(v)$

Lemma. Every Markov chain P has at least one stationary distribution

Proof.

- $\boldsymbol{P}$ is stochastic $\implies \boldsymbol{P}\mathbf{1} = \mathbf{1} \implies 1$ is an eigenvalue of $\boldsymbol{P} \implies 1$ is an eigenvalue of $\boldsymbol{P}^\top$
- $\exists \boldsymbol{v} \in \mathbb{R}^\Omega$ s.t. $\boldsymbol{v}\boldsymbol{P} = \boldsymbol{v}$
- For any $x \in \Omega$, $|\boldsymbol{v}(x)| = \left|\sum_{y\in\Omega} \boldsymbol{v}(y)\boldsymbol{P}(y \to x)\right| \leq \sum_{y\in\Omega} |\boldsymbol{v}(y)|\boldsymbol{P}(y \to x)$
- If $|\boldsymbol{v}(x)| < \sum_{y\in\Omega} |\boldsymbol{v}(y)|\boldsymbol{P}(y \to x)$, then summing over x gives:

$$\sum_{x\in\Omega} |\boldsymbol{v}(x)| < \sum_{x\in\Omega}\sum_{y\in\Omega} |\boldsymbol{v}(y)|\boldsymbol{P}(y \to x) = \sum_{y\in\Omega} |\boldsymbol{v}(y)| \sum_{x\in\Omega} \boldsymbol{P}(y \to x) = \sum_{y\in\Omega} |\boldsymbol{v}(y)|$$

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Markov Chains

A probability distribution μ on Ω is stationary with respect to a Markov chain P if $\mu \boldsymbol{P} = \mu$

- For the random walk on a graph G, the stationary distribution $\mu_G(v) \propto \deg_G(v)$

Lemma. Every Markov chain P has at least one stationary distribution

Lemma. If a Markov chain P is reversible w.r.t. μ, then μ is a stationary distribution

Proof.

$$\mu \boldsymbol{P}(x) = \sum_y \mu(y) \boldsymbol{P}(y \to x) = \sum_y \mu(x) \boldsymbol{P}(x \to y) = \mu(x)$$

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Markov Chains

We say a Markov chain P is ergodic if P satisfies both of the following properties:

- Irreducibility: P is irreducible if for all $x, y \in \Omega$, there exists $t \geq 0$ such that $\boldsymbol{P}^t(x \to y) > 0$
- Aperiodicity: The period of a state $x \in \Omega$ under P is defined as the greatest common divisor (gcd) of $\{t \geq 1 : \boldsymbol{P}^t(x, x) > 0\}$. We say P is aperiodic if all states have period 1

- Irreducibility means the underlying weighted directed graph of P is strongly connected
- Aperiodicity is easy to satisfy by the lazyfication of P: we can let $\boldsymbol{P}' := \frac{\boldsymbol{I}+\boldsymbol{P}}{2}$. That is, in each step, there is a $\frac{1}{2}$-probability of staying in the same state

Lemma. Let P be a Markov chain on a finite state space Ω. If P is irreducible and aperiodic, then there exists $t_\star$ such that for all $x, y \in \Omega$, $\boldsymbol{P}^{t_\star}(x \to y) > 0$

Markov Chains

Theorem (Fundamental Theorem of Markov Chains**).** Let P be an ergodic Markov chain on a state space Ω. Then P has a unique stationary distribution μ on Ω. Furthermore, for every initial distribution μ_0, the distribution $\mu_t \coloneqq \mu_0 \boldsymbol{P}^t$ of X_t converges (pointwise) to μ as $t \to \infty$

Proof.

- We will only prove the theorem for reversible Markov chains. See e.g. (Levin-Peres-Wilmer '17) for the general case
- If P is reversible w.r.t. μ, then P is similar to a symmetric matrix under transformation by the diagonal matrix $\boldsymbol{D} \coloneqq \mathbf{Diag}\left(\sqrt{\mu(x)}\right)$:

$$\boldsymbol{DPD}^{-1}(x,y) = \sqrt{\mu(x)}P(x \to y)\frac{1}{\sqrt{\mu(y)}} = \sqrt{\mu(y)}P(y \to x)\frac{1}{\sqrt{\mu(x)}} = \boldsymbol{DPD}^{-1}(y,x)$$

- Thus, P is diagonalizable with real eigenvalues and eigenvectors

Markov Chains

Theorem (Fundamental Theorem of Markov Chains). Let P be an ergodic Markov chain on a state space Ω. Then P has a unique stationary distribution μ on Ω. Furthermore, for every initial distribution μ_0, the distribution $\mu_t := \mu_0 \boldsymbol{P}^t$ of X_t converges (pointwise) to μ as $t \to \infty$

Proof.

Theorem (Perron-Frobenius). Any irreducible, aperiodic stochastic matrix $\boldsymbol{P}$ has an eigenvalue $\lambda_1 = 1$ with unique associated left eigenvector $\boldsymbol{v}_1 > 0$. Moreover, all other eigenvalues λ_i satisfy $|\lambda_i| < 1$

- We know that if P is reversible w.r.t. μ, then P has real eigenvalues and μ is a stationary distribution, i.e., an eigenvector with eigenvalue 1
- Perron-Frobenius theorem implies that μ is the unique stationary distribution
- The convergence follows from a similar proof as the Power Method

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Markov Chains

Theorem (Fundamental Theorem of Markov Chains**).** Let P be an ergodic Markov chain on a state space Ω. Then P has a unique stationary distribution μ on Ω. Furthermore, for every initial distribution μ_0, the distribution $\mu_t \coloneqq \mu_0 \boldsymbol{P}^t$ of X_t converges (pointwise) to μ as $t \to \infty$

- $\mu_t = \mu_0 P^t = \langle \mu_0, \boldsymbol{v}_1 \rangle \boldsymbol{v}_1 + \sum_{i>1} \langle \mu_0, \boldsymbol{v}_i \rangle \lambda_i^t \boldsymbol{v}_i$
- The convergence rate depends on $\lambda_1 - \lambda_2 = 1 - \lambda_2$, the spectral gap of $\boldsymbol{P}$
  - We can make $\lambda_{|\Omega|}$ be bounded away from -1 by lazyfication
- If P is irreducible but not aperiodic, then the Markov chain may not converge to μ from every starting state:

$$\boldsymbol{P} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \mu = [1/2 \quad 1/2], \qquad \lambda_1 = 1, \lambda_2 = -1$$

Today's Lecture

- Introduction to Markov Chains
- Speed of Convergence
- Spectral Theory of Markov Chains
- Poincaré Inequality

Speed of Convergence

Fix an ergodic Markov chain P with stationary distribution μ on a state space Ω. Let $\epsilon > 0$ be an error parameter. We define the ϵ-mixing time of P (with initial distribution μ_0) by

$$T_{\mathrm{mix}}(\epsilon; \mu_0, P) \coloneqq \min\{t \geq 0 : \|\mu_0 P^t - \mu\|_{\mathrm{TV}} \leq \epsilon\}$$
$$T_{\mathrm{mix}}(\epsilon;\ P) \coloneqq \sup_{\mu_0} T_{\mathrm{mix}}(\epsilon; \mu_0, P)$$

We define the total variation mixing time of P to be $T_{\mathrm{mix}} \coloneqq T_{\mathrm{mix}}(1/4)$

- The constant $1/4$ is arbitrary. It holds that $T_{\mathrm{mix}}(\epsilon) \lesssim T_{\mathrm{mix}} \cdot \log(1/\epsilon)$ (Levin-Peres-Wilmer '17)
- We can define a version of mixing time with respect to any metric on probability measures (e.g. Wasserstein distance, KL-divergence, χ^2-divergence, etc.)
- Fast mixing: T_{mix} is polynomial in the system size
- Rapid mixing: T_{mix} is logarithmic in the system size

Detour: Metrics and Divergences between Probability Distributions

Let μ_1, μ_2 be two probability distributions

- Total variation distance

$$\|\mu_1 - \mu_2\|_{\mathrm{TV}} \coloneqq \sup_{S \subset \Omega} \mu_1(S) - \mu_2(S) = \frac{1}{2} \int \left| \frac{d\mu_1}{dx} - \frac{d\mu_2}{dx} \right| dx = \frac{1}{2} \|\mu_1 - \mu_2\|_1$$
$$= \sup_{f:\Omega \to \mathbb{R}: \|f\|_\infty \leq 1} \left| \mathbb{E}_{\mu_1}[f] - \mathbb{E}_{\mu_2}[f] \right|$$

- Wasserstein distance

$$\|\mu_1 - \mu_2\|_{W_1} \coloneqq \sup_{f:\Omega \to \mathbb{R}: 1-\mathrm{Lip}} \left| \mathbb{E}_{\mu_1}[f] - \mathbb{E}_{\mu_2}[f] \right|$$

$$\|\mu_1 - \mu_2\|_{W_p} \coloneqq \inf_{\pi \in \Gamma(\mu_1, \mu_2)} \mathbb{E}_{(X,Y)\sim\pi}\left[\|X - Y\|_2^p\right]^{1/p}, \qquad \Gamma(\mu_1, \mu_2) \coloneqq \{\text{couplings of } \mu_1 \text{ and } \mu_2\}$$

Detour: Metrics and Divergences between Probability Distributions

Let μ_1, μ_2 be two probability distributions

- Kullback-Leibler divergence

$$D_{\mathrm{KL}}(\mu_1 \| \mu_2) \coloneqq \sum_x \mu_1(x) \ln \frac{\mu_1(x)}{\mu_2(x)}$$

- Chi-square divergence

$$\chi^2(\mu_1 \| \mu_2) \coloneqq \sum_x \frac{\left(\mu_1(x) - \mu_2(x)\right)^2}{\mu_2(x)}$$

- Hellinger distance

$$\|\mu_1 - \mu_2\|_{\mathrm{H}} \coloneqq \|\sqrt{\mu_1} - \sqrt{\mu_2}\|_2$$

Detour: Metrics and Divergences between Probability Distributions

$$\frac{1}{2}\|\mu_1-\mu_2\|_{\mathrm{H}}^2 \le \|\mu_1-\mu_2\|_{\mathrm{TV}} \le \|\mu_1-\mu_2\|_{\mathrm{H}} \le \sqrt{D_{\mathrm{KL}}(\mu_1\|\mu_2)} \le \sqrt{\chi^2(\mu_1\|\mu_2)}$$

$$\|\mu_1-\mu_2\|_{\mathrm{TV}} \le \frac{1}{2}\sqrt{\chi^2(\mu_1\|\mu_2)}$$

$$\|\mu_1-\mu_2\|_{\mathrm{TV}} \le \sqrt{D_{\mathrm{KL}}(\mu_1\|\mu_2)/2}$$

(Pinsker's Inequality)

$$\|\mu_1-\mu_2\|_{\mathrm{TV}} \le \sqrt{1-e^{-D_{\mathrm{KL}}(\mu_1\|\mu_2)}} \le 1-\frac{1}{2}e^{-D_{\mathrm{KL}}(\mu_1\|\mu_2)}$$

(Bretagnolles–Huber Bound)

$$D_{\mathrm{KL}}(\mu_1\|\mu_2) \le \ln\left(1+\chi^2(\mu_1\|\mu_2)\right) \le \chi^2(\mu_1\|\mu_2)$$

$$\|\mu_1-\mu_2\|_{W_1} \le \mathrm{diam}(\Omega)\cdot\|\mu_1-\mu_2\|_{\mathrm{TV}}$$

If Ω finite, $d_{\min}\cdot\|\mu_1-\mu_2\|_{\mathrm{TV}} \le \|\mu_1-\mu_2\|_{W_1}$, where $d_{\min} \coloneqq \min_{x\neq y}\|x-y\|_2$

Speed of Convergence

Lemma (Data processing inequality**).** Let μ, ν be two probability measures on a finite state space Ω, and let P be a Markov chain on Ω. Then

$$\|\mu \boldsymbol{P} - \nu \boldsymbol{P}\|_{\mathrm{TV}} \leq \|\mu - \nu\|_{\mathrm{TV}}$$

- Let μ be stationary w.r.t. P
- Then, applying one step of P to any other distribution ν cannot increase its total variation distance from μ
- So, P cannot "unmix" a distribution

Today's Lecture

- Introduction to Markov Chains
- Speed of Convergence
- Spectral Theory of Markov Chains
- Poincaré Inequality

Spectral Theory of Markov Chains

Let μ be a probability distribution over a finite state space Ω. This distribution induces an inner product on the space of functions $\{f: \Omega \to \mathbb{R}\} \cong \mathbb{R}^{\Omega}$ given by

$$\langle f, g \rangle_\mu \coloneqq \mathbb{E}_\mu[fg] = \sum_{x \in \Omega} \mu(x) f(x) g(x)$$

- We have shown that if a Markov chain P is reversible, then $\boldsymbol{P}$ has real eigenvalues and eigenvectors by an explicit diagonalization. Here, we give an alternative, more abstract proof

Lemma. A Markov chain P is reversible w.r.t. μ if and only if $\boldsymbol{P}$ is self-adjoint w.r.t. $\langle \cdot, \cdot \rangle_\mu$, i.e.

$$\langle f, \boldsymbol{P} g \rangle_\mu = \langle \boldsymbol{P} f, g \rangle_\mu \qquad \forall f, g: \Omega \to \mathbb{R}$$

- For any inner product $(\cdot,\cdot): \mathcal{H} \times \mathcal{H} \to \mathbb{R}$ on some space $\mathcal{H}$, and for any operator A, the adjoint of A is defined as the unique linear operator A^* such that

$$(f, Ag) = (A^* f, g) \qquad \forall f, g \in \mathcal{H}$$

- If $\mathcal{H}$ is a finite dimensional real vector space, then $A^* = A^\top$, the transpose of the matrix

Spectral Theory of Markov Chains

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$$\langle f, \boldsymbol{P} g \rangle_{\mu} = \langle \boldsymbol{P} f, g \rangle_{\mu} \qquad \forall f, g: \Omega \to \mathbb{R}$$

Theorem (General spectral theorem). Let A be an $m \times m$ matrix which is self-adjoint w.r.t. $(\cdot, \cdot)$. Then A has m real eigenvalues $\lambda_1, \dots, \lambda_m$ and m eigenvectors $u_1, \dots, u_m$ which are $(\cdot, \cdot)$–orthonormal:

$$(u_i, u_j) = \delta_{ij} \qquad \forall i, j \in [m]$$

Spectral Theory of Markov Chains

Spectral gap of a Markov chain

- We can order the eigenvalues of $\boldsymbol{P}$ as $-1 \leq \lambda_{|\Omega|} \leq \cdots \leq \lambda_2 \leq \lambda_1 = 1$
- Let $\lambda_\star \coloneqq \max_{i>1} |\lambda_i| = \max\{\lambda_2, |\lambda_{|\Omega|}|\}$.
- Irreducibility $\Longrightarrow \lambda_2 < 1$ (Perron-Frobenius)
- Aperiodicity $\Longrightarrow \lambda_{|\Omega|} > -1$
- Thus, ergodicity $\Longrightarrow \lambda_\star < 1$
- Spectral gap $\gamma \coloneqq 1 - \lambda_2$, and absolute spectral gap $\gamma_\star \coloneqq 1 - \lambda_\star$
- After lazyfication, $\lambda_i \geq 0 \; \forall i \in [|\Omega|]$ and $\gamma_\star = \gamma$

Spectral Theory of Markov Chains

Theorem (Spectral gap implies rapid mixing). Let P be an ergodic Markov chain on Ω which is reversible w.r.t. μ. Then for every $x \in \Omega$ and every $\epsilon > 0$,

$$T_{\mathrm{mix}}(\epsilon; \delta_x, P) \leq \frac{1}{\gamma_\star} \ln\left(\frac{1}{2\epsilon\sqrt{\mu(x)}}\right)$$

In particular,

$$T_{\mathrm{mix}}(\epsilon) \leq \frac{1}{\gamma_\star}\left(\frac{1}{2}\ln\frac{1}{\mu_{\min}} + \ln\frac{1}{2\epsilon}\right)$$

where $\mu_{\min} := \min_{x\in\Omega} \mu(x)$ is the minimum nonzero probability under μ

Spectral Theory of Markov Chains

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Theorem (Rapid mixing implies spectral gap). Let P be an ergodic Markov chain on Ω which is reversible w.r.t. μ. Then for every $x \in \Omega$ and every $\epsilon > 0$,

$$T_{\mathrm{mix}}(\epsilon) \geq \left(\frac{1}{\gamma_\star} - 1\right)\log\frac{1}{2\epsilon}$$

Spectral Theory of Markov Chains

Lemma (Spectral Decomposition of Reversible Markov Chains**).** Let P be reversible w.r.t. μ. Then $\mathbb{R}^{\Omega}$ admits a basis $\{\psi_i\}_{i=1}^{|\Omega|}$ satisfying the following:

1. Each ψ_i is a (right) eigenfunction of P, i.e. $\boldsymbol{P}\psi_i = \lambda_i \psi_i$. Furthermore, $\psi_1 = \mathbf{1}$
2. The collection $\{\psi_i\}_{i=1}^{|\Omega|}$ is orthonormal w.r.t. the inner product $\langle \cdot, \cdot \rangle_{\mu}$
3. For every $x, y \in \Omega$ and every $t \in \mathbb{N}$,

$$\frac{\boldsymbol{P}^t(x \to y)}{\mu(y)} = \sum_{i=1}^{|\Omega|} \lambda_i^t \psi_i(x) \psi_i(y)$$

In particular, we may express the measure $\delta_x \boldsymbol{P}^t$ in density form as

$$\frac{d(\delta_x \boldsymbol{P}^t)}{d\mu} = \sum_{i=1}^{|\Omega|} \lambda_i^t \psi_i(x) \cdot \psi_i$$

Spectral Theory of Markov Chains

Proof.

- We know that $\boldsymbol{A}_\mu \coloneqq \boldsymbol{D}_\mu^{1/2} \cdot \boldsymbol{P} \cdot \boldsymbol{D}_\mu^{-1/2}$ with $\boldsymbol{D}_\mu = \mathbf{Diag}(\mu)$ is a real symmetric matrix
- Let $\boldsymbol{A}_\mu = \sum_{i=1}^{|\Omega|} \lambda_i \varphi_i \varphi_i^\top$ be its eigendecomposition
- Note that $\boldsymbol{A}_\mu \sqrt{\mu} = \boldsymbol{D}_\mu^{1/2} \cdot \boldsymbol{P} \cdot \mathbf{1} = \boldsymbol{D}_\mu^{1/2} \cdot \mathbf{1} = \sqrt{\mu}$. Thus, we can let $\varphi_1 \coloneqq \sqrt{\mu}$
- We further observe that $\boldsymbol{P} = \sum_{i=1}^{|\Omega|} \lambda_i \boldsymbol{D}_\mu^{-1/2} \varphi_i \varphi_i^\top \boldsymbol{D}_\mu^{1/2}$, and

$$\boldsymbol{P} \cdot \boldsymbol{D}_\mu^{-1/2} \varphi_i = \sum_{j=1}^{|\Omega|} \lambda_j \boldsymbol{D}_\mu^{-1/2} \varphi_j \varphi_j^\top \boldsymbol{D}_\mu^{1/2} \cdot \boldsymbol{D}_\mu^{-1/2} \varphi_i = \lambda_i \boldsymbol{D}_\mu^{-1/2} \varphi_i$$

- Let $\psi_i \coloneqq \boldsymbol{D}_\mu^{-1/2} \varphi_i$. We have $\psi_1 = \boldsymbol{D}_\mu^{-1/2} \sqrt{\mu} = \mathbf{1}$, and

$$\langle \psi_i, \psi_j \rangle_\mu = \left(\boldsymbol{D}_\mu^{-1/2} \varphi_i \right)^\top \boldsymbol{D}_\mu \left(\boldsymbol{D}_\mu^{-1/2} \varphi_j \right) = \langle \varphi_i, \varphi_j \rangle$$

Spectral Theory of Markov Chains

Proof.

- For any $t \in \mathbb{N}$, we have

$$\boldsymbol{P}^t = \left(\boldsymbol{D}_\mu^{-1/2} \cdot \boldsymbol{A}_\mu \cdot \boldsymbol{D}_\mu^{1/2}\right)^t = \boldsymbol{D}_\mu^{-1/2} \cdot \boldsymbol{A}_\mu^t \cdot \boldsymbol{D}_\mu^{1/2} = \boldsymbol{D}_\mu^{-1/2} \sum_{i=1}^{|\Omega|} \lambda_i^t \varphi_i \varphi_i^\top \boldsymbol{D}_\mu^{1/2}$$

$$= \boldsymbol{D}_\mu^{-1/2} \sum_{i=1}^{|\Omega|} \lambda_i^t \left(\boldsymbol{D}_\mu^{1/2} \psi_i\right)\left(\psi_i^\top \boldsymbol{D}_\mu^{1/2}\right) \boldsymbol{D}_\mu^{1/2} = \sum_{i=1}^{|\Omega|} \lambda_i^t \psi_i \psi_i^\top \boldsymbol{D}_\mu$$

- Thus, for any $x, y \in \Omega$,

$$\boldsymbol{P}^t(x \to y) = \sum_{i=1}^{|\Omega|} \lambda_i^t \psi_i(x) \psi_i(y) \mu(y)\,, \qquad \frac{d(\delta_x \boldsymbol{P}^t)}{d\mu} = \sum_{i=1}^{|\Omega|} \lambda_i^t \psi_i(x) \psi_i^\top$$

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Spectral Theory of Markov Chains

- To prove the mixing time in total variation distance, we use χ^2-divergence as a bridge:

$$\chi^2(\mu\|\nu) = \sum_x \frac{\big(\mu(x) - \nu(x)\big)^2}{\nu(x)} = \sum_x \left(\frac{\mu(x)^2}{\nu(x)} - 2\mu(x) + \nu(x)\right)$$

$$= \mathbb{E}_\nu\left[\left(\frac{d\mu}{d\nu}\right)^2\right] - 1 = \mathrm{Var}_\nu\left[\frac{d\mu}{d\nu}\right]$$

- χ^2-divergence can upper bound TV distance:

$$\|\mu - \nu\|_{\mathrm{TV}}^2 \le \frac{1}{4}\chi^2(\mu\|\nu) = \frac{1}{4}\mathrm{Var}_\nu\left[\frac{d\mu}{d\nu}\right]$$

- To upper bound $T_{\mathrm{mix}}(\epsilon; \delta_x, P)$, it suffices to upper bound $\chi^2(\delta_x \boldsymbol{P}^t\|\mu)$

Spectral Theory of Markov Chains

Proof of Spectral gap implies rapid mixing:

- We bound the χ^2-divergence between $\delta_x \boldsymbol{P}^t$ and μ:

$$
\begin{aligned}
\mathrm{Var}_\mu\left[\frac{d(\delta_x \boldsymbol{P}^t)}{d\mu}\right] &= \mathbb{E}_\mu\left[\left(\sum_{i=1}^{|\Omega|} \lambda_i^t \psi_i(x) \cdot \psi_i\right)^2\right] - 1 \\
&= \sum_{i,j=1}^{|\Omega|} \lambda_i^t \lambda_j^t \psi_i(x)\psi_j(x) \cdot \langle \psi_i, \psi_j \rangle_\mu - 1 \\
&= \sum_{i=1}^{|\Omega|} \lambda_i^{2t} \psi_i(x)^2 - 1 = \sum_{i=2}^{|\Omega|} \lambda_i^{2t} \psi_i(x)^2 \\
&\leq \lambda_\star^{2t} \sum_{i=2}^{|\Omega|} \psi_i(x)^2 = \frac{\lambda_\star^{2t}}{\mu(x)^2} \sum_{i=2}^{|\Omega|} \langle \psi_i, \delta_x \rangle_\mu^2 \\
&\leq \frac{\lambda_\star^{2t}}{\mu(x)^2} \langle \delta_x, \delta_x \rangle_\mu^2 = \frac{\lambda_\star^{2t}}{\mu(x)}
\end{aligned}
$$

Spectral Theory of Markov Chains

Proof of Spectral gap implies rapid mixing:

- $\chi^2(\delta_x \boldsymbol{P}^t \| \mu) \leq \frac{\lambda_\star^{2t}}{\mu(x)}$
- Thus, $\|\delta_x \boldsymbol{P}^t - \mu\|_{\mathrm{TV}} \leq \frac{\lambda_\star^t}{2\sqrt{\mu(x)}}$, which gives

$$\frac{\lambda_\star^t}{2\sqrt{\mu(x)}} \leq \epsilon \Leftrightarrow \left(\frac{1}{\lambda_\star}\right)^t \geq \frac{1}{2\epsilon\sqrt{\mu(x)}} \Leftrightarrow t \geq \ln\left(\frac{1}{2\epsilon\sqrt{\mu(x)}}\right) \Big/ \ln\left(\frac{1}{\lambda_\star}\right)$$

- $\ln\left(\frac{1}{\lambda_\star}\right) = \ln\left(\frac{1}{1-\gamma_\star}\right) \geq \gamma_\star$
- Thus, $t \geq \frac{1}{\gamma_\star} \ln\left(\frac{1}{2\epsilon\sqrt{\mu(x)}}\right)$ suffices to ensure the TV distance $\leq \epsilon$
- That is, $T_{\mathrm{mix}}(\epsilon; \delta_x, P) \leq \frac{1}{\gamma_\star} \ln\left(\frac{1}{2\epsilon\sqrt{\mu(x)}}\right)$

■

Spectral Theory of Markov Chains

Proof of Rapid mixing implies spectral gap:

- Let ψ be the eigenfunction of P with eigenvalue $\lambda_\star$. Then, $\psi \neq \psi_1$ and $\langle \psi, \mathbf{1} \rangle_\mu = 0$
- For any $t \in \mathbb{N}$ and $x \in \Omega$, we have

$$
\begin{aligned}
|\boldsymbol{P}^t \psi(x)| &= \lambda_\star^t \cdot |\psi(x)| \\
&= \left| \sum_{y \in \Omega} \boldsymbol{P}^t(x \to y) \psi(y) \right| \\
&= \left| \sum_{y \in \Omega} \boldsymbol{P}^t(x \to y) \psi(y) - \langle \psi, \mathbf{1} \rangle_\mu \right| = \left| \sum_{y \in \Omega} \big(\boldsymbol{P}^t(x \to y) - \mu(y) \big) \psi(y) \right| \\
&\leq \max_{y \in \Omega} |\psi(y)| \cdot \sum_{y \in \Omega} |\boldsymbol{P}^t(x \to y) - \mu(y)| = 2 \max_{y \in \Omega} |\psi(y)| \cdot \|\delta_x \boldsymbol{P}^t - \mu\|_{\mathrm{TV}}
\end{aligned}
$$

- Thus, $\lambda_\star^t \leq 2 \max_{x \in \Omega} \|\delta_x \boldsymbol{P}^t - \mu\|_{\mathrm{TV}} \leq 2\epsilon$ if $t = T_{\mathrm{mix}}(\epsilon)$

Spectral Theory of Markov Chains

Proof of Rapid mixing implies spectral gap:

- Thus, $\lambda_\star^t \leq 2 \max_{x \in \Omega} \|\delta_x \boldsymbol{P}^t - \mu\|_{\mathrm{TV}} \leq 2\epsilon$ if $t = T_{\mathrm{mix}}(\epsilon)$
- We get that

$$T_{\mathrm{mix}}(\epsilon) \geq \ln\left(\frac{1}{2\epsilon}\right) \Big/ \ln\left(\frac{1}{\lambda_\star}\right) \geq \ln\left(\frac{1}{2\epsilon}\right) \Big/ \left(\frac{1}{\lambda_\star} - 1\right) = \left(\frac{1}{\gamma_\star} - 1\right) \ln\left(\frac{1}{2\epsilon}\right)$$

■

Today's Lecture

- Introduction to Markov Chains
- Speed of Convergence
- Spectral Theory of Markov Chains
- Poincaré Inequality

Poincaré Inequality

Define the Dirichlet form of a reversible Markov chain P by

$$\mathcal{E}_P(f,g) \coloneqq \langle f, (\boldsymbol{I}-\boldsymbol{P})g\rangle_\mu$$
$$= \frac{1}{2}\sum_{x,y\in\Omega} \mu(x)\boldsymbol{P}(x\to y)\big(f(x)-f(y)\big)\big(g(x)-g(y)\big)$$

- When $f=g$, $\mathcal{E}_P(f,f) = \frac{1}{2}\mathbb{E}_{x\sim\mu, y\sim\boldsymbol{P}(x\to\cdot)}\left[\big(f(x)-f(y)\big)^2\right]$ can be recognized as a measure of "local variance", since $\mathrm{Var}_\mu[f] = \frac{1}{2}\mathbb{E}_{x,y\sim\mu}\left[\big(f(x)-f(y)\big)^2\right]$

We say P satisfies a Poincaré Inequality with constant $C>0$ if

$$C\cdot \mathrm{Var}_\mu[f] \le \mathcal{E}_P(f,f) \qquad \forall f:\Omega\to\mathbb{R}$$

- C is called the Poincaré constant of P

Poincaré Inequality

Lemma. Let P be a reversible Markov chain. Then we have the identity

$$\gamma = \inf_f \frac{\mathcal{E}_P(f,f)}{\mathrm{Var}_\mu[f]} = \inf_{f:\langle f,\mathbf{1}\rangle_\mu=0} \frac{\langle f,(\boldsymbol{I}-\boldsymbol{P})f\rangle_\mu}{\langle f,f\rangle_\mu}$$

where the infimum is over all functions with nonzero global variance

- This is just the Courant–Fischer–Weyl min-max principle for $\boldsymbol{I}-\boldsymbol{P}$
- It implies that the Poincaré constant of P is the spectral gap γ

Poincaré Inequality

Lemma. Let P be reversible w.r.t. μ. Then, for any ν,

$$\chi^2(\nu \boldsymbol{P} \| \mu) \leq \lambda_\star^2 \cdot \chi^2(\nu \| \mu)$$

Proof.

- First note that for any $g: \Omega \to \mathbb{R}$, $\left\langle \frac{d(\nu \boldsymbol{P})}{d\mu}, g \right\rangle_\mu = \nu^\top \boldsymbol{P} g = \left\langle \frac{d\nu}{d\mu}, \boldsymbol{P} g \right\rangle_\mu = \left\langle \boldsymbol{P} \frac{d\nu}{d\mu}, g \right\rangle_\mu$
- Recall that $\chi^2(\nu \| \mu) = \mathrm{Var}_\mu \left[\frac{d\nu}{d\mu}\right]$. We have

$$\mathrm{Var}_\mu \left[\frac{d\nu}{d\mu}\right] - \mathrm{Var}_\mu \left[\frac{d(\nu \boldsymbol{P})}{d\mu}\right] = \left\langle \frac{d\nu}{d\mu}, \frac{d\nu}{d\mu} \right\rangle_\mu - \left\langle \boldsymbol{P} \frac{d\nu}{d\mu}, \boldsymbol{P} \frac{d\nu}{d\mu} \right\rangle_\mu = \left\langle \frac{d\nu}{d\mu}, (\boldsymbol{I} - \boldsymbol{P}^2) \frac{d\nu}{d\mu} \right\rangle_\mu = \mathcal{E}_{\boldsymbol{P}^2}\left(\frac{d\nu}{d\mu}, \frac{d\nu}{d\mu}\right)$$

$$\text{(PI for } P^2\text{)} \geq (1 - \lambda_\star^2) \mathrm{Var}_\mu \left[\frac{d\nu}{d\mu}\right]$$

■

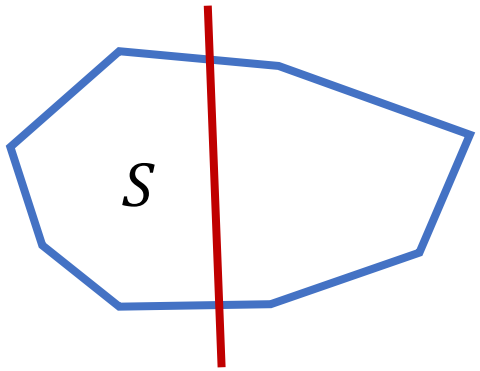
Poincaré Inequality

Let P be a Markov chain which is reversible w.r.t. a distribution μ on Ω. For every $S \subseteq \Omega$, define the conductance of S w.r.t. P to be the ratio

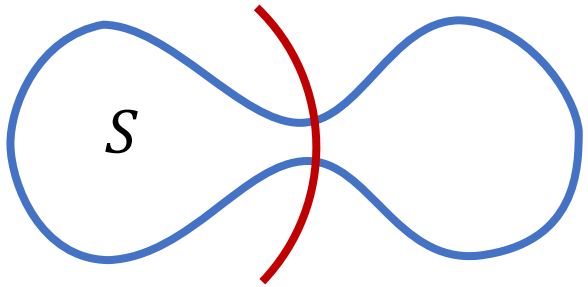
$$\Phi(S) := \frac{\sum_{x \in S, y \in \Omega \setminus S} \mu(x) P(x \to y)}{\sum_{x \in S} \mu(x)}$$

The conductance of P is

$$\Phi(P) := \sup_{S \subset \Omega : \mu(S) \leq 1/2} \Phi(S)$$



$\Phi(P)$ large, no bottleneck, mix fast



$\Phi(P)$ small, trapped within S, mix slowly

Poincaré Inequality

Theorem (Cheeger's Inequality; Sinclair-Jerrum '89, Lawler-Sokal '88**).** For every reversible Markov chain P,

$$\frac{1}{2}\Phi(P)^2 \leq \gamma \leq 2\Phi(P)$$

- $T_{\text{mix}} = \mathcal{O}\left(\Phi(P)^2 \ln \mu_{\min}^{-1}\right)$
- Upper bound is easy: apply Poincaré Inequality $\gamma \cdot \text{Var}_\mu[f] \leq \mathcal{E}_P(f,f)$ with $f = \mathbf{1}_S$:
  - $\text{Var}_\mu[f] = \mathbb{E}_\mu[\mathbf{1}_S^2] - \mathbb{E}_\mu[\mathbf{1}_S]^2 = \mu(S) - \mu(S)^2$
  - $\mathcal{E}_P(f,f) = \frac{1}{2}\sum_{x,y\in\Omega} \mu(x)\boldsymbol{P}(x \to y)\left(\mathbf{1}_S(x) - \mathbf{1}_S(y)\right)^2 = \frac{1}{2}\sum_{x\in S, y\in\Omega\setminus S}\left(\mu(x)P(x \to y) + \mu(y)P(y \to x)\right) = \sum_{x\in S, y\in\Omega\setminus S} \mu(x)P(x \to y)$
  - $\gamma \leq \frac{\mathcal{E}_P(f,f)}{\text{Var}_\mu[f]} = \frac{\sum_{x\in S, y\in\Omega\setminus S} \mu(x)P(x\to y)}{\mu(S)-\mu(S)^2} \leq 2\frac{\sum_{x\in S, y\in\Omega\setminus S} \mu(x)P(x\to y)}{\mu(S)} \leq 2\Phi(P)$ for any S with $\mu(S) \leq \frac{1}{2}$
- Lower bound is non-trivial

Poincaré Inequality

- The conductance is still hard to use, since it needs to consider every cut
- An easier approach to estimate the Poincaré constant is the Canonical Path method, which is the dual picture of the conductance
- The canonical path method considers a multicommodity flow problem on the state space graph of the Markov chain

Poincaré Inequality

- An ergodic Markov chain P with stationary distribution π $\longleftrightarrow$ a weighted graph $G = (\Omega, E)$
- For every $e = (x, y) \in \Omega \times \Omega$, we define the capacity of this edge to be $C(e) \coloneqq \pi(x)\boldsymbol{P}(x \to y)$, the ergodic flow from x to y
- For every $(x, y) \in \Omega \times \Omega$, we define the demand $D(x, y) \coloneqq \mu(x)\mu(y)$
- A flow f is a scheme that routes $D(x, y)$ "units" of flow from x to y simultaneously for all $x, y \in \Omega$
  - $f: \mathcal{P} \to \mathbb{R}_{\geq 0}$, where $\mathcal{P} = \bigcup_{xy} \mathcal{P}_{xy}$ and $\mathcal{P}_{xy}$ is the set of simple paths from x to y
  - $\sum_{p \in \mathcal{P}_{xy}} f(p) = D(x, y)$ for all $x, y \in \Omega$
- The total flow through an edge e is $f(e) = \sum_{p \in \mathcal{P}: e \in p} f(p)$
- The congestion (or cost) of an edge e is $\rho(e) = \frac{f(e)}{C(e)}$, and the congestion of the flow f is $\rho(f) = \max_e \rho(e)$
- The length $\ell(f)$ of f is $\ell(f) \coloneqq \max_{p: f(p) > 0} |p|$

Poincaré Inequality

Theorem (Jerrum–Sinclair ’89, Diaconis-Stroock ’91**).** For any ergodic P and any flow f for P, let ℓ the Poincaré constant γ is bounded by

$$\gamma \geq \frac{1}{\rho(f)\ell(f)}$$

- $T_{\text{mix}} = \mathcal{O}\left(\rho(f)\ell(f)\ln \mu_{\min}^{-1}\right)$
- The strategy to bound mixing time is to construct any set of short paths $\mathcal{P}$ with small congestions

Poincaré Inequality

- Consider lazy random walk on the hypercube $\{0,1\}^n$: for every state $x \in \{0,1\}^n$, with probability $1/2$, stay at x; and with probability $1/2$, flip a randomly chosen coordinate
- $\mu(x) = \frac{1}{2^n}$ (uniform distribution), $C(x,y) = \frac{1}{2^n \cdot 2n} = \frac{1}{2^{n+1}n}$ for every $x \neq y \in \{0,1\}^n$, $D(x,y) = 2^{-2n}$
- Intuitively, to maximize $\frac{1}{\rho(f)\ell(f)}$, we should spread the flow between x and y evenly among all shortest paths from x to y
- To compute $\rho(f)$, we need to know $f(e)$ for every edge e. By the symmetry of the hypercube, every edge should receive the same amount of flow

$$f(e) = \frac{\sum_{e \in E} f(e)}{|E|} = \frac{\sum_{xy} D(x,y)\mathrm{dist}(x,y)}{2^n n} = \frac{2^{-2n}\sum_{xy}\mathrm{dist}(x,y)}{2^n n} = \frac{2^{-2n} \cdot 2^{2n} n/2}{2^n n} = 2^{-(n+1)}$$

- $\rho(f) = 2^{-(n+1)}/\left(2^{-(n+1)}n^{-1}\right) = n$ and $\ell(f) = n$

The average distance is $n/2$

- $T_{\mathrm{mix}} = \mathcal{O}\left(n^2 \ln \mu_{\min}^{-1}\right) = \mathcal{O}(n^3)$, which is not tight (it can be improved to $\mathcal{O}(n \log n)$)